

The empirical recurrence for OEIS A316286, recovered from its tail

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8 September 2026

Abstract

OEIS A316286 records “Empirical recurrence of order 85” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 137$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 5$, so the recurrence is proved for every $n \geq 90$. Its last coefficient is $c_{85} = 13312 \neq 0$, so no further tail satisfies anything shorter; any order-85 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture that is not written down

OEIS A316286 is “Number of $n \times 5$ 0..1 arrays with every element unequal to 0, 1, 2, 3, 6 or 8 king-move adjacent elements, with upper left element zero.”. It has offset 1 and begins

16, 120, 361, 1372, 7973, 35706, 164734, 838632, ...

The entry states:

Empirical recurrence of order 85 (see link above).

As of the “Last modified” line on the live entry (Jun 16 2023 15:27:45, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 137$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 137$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 85: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 5$ is the first whose minimal order is exactly the stated 85.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 274$ exact terms from $n = 5$ on, it returns order 85 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{85} c_i a(n-i),$$

c_1	11	c_{23}	594277991	c_{45}	−7544421284	c_{67}	−158327520
c_2	−38	c_{24}	−550717101	c_{46}	7511954341	c_{68}	282603370
c_3	104	c_{25}	6477836	c_{47}	1628169512	c_{69}	−209090044
c_4	−690	c_{26}	−1266599980	c_{48}	−1762341726	c_{70}	136377156
c_5	2263	c_{27}	1112624851	c_{49}	5115424610	c_{71}	−141558568
c_6	−3612	c_{28}	636384478	c_{50}	−9820768861	c_{72}	83566816
c_7	17141	c_{29}	1781022464	c_{51}	9176183140	c_{73}	−46006904
c_8	−57033	c_{30}	−1263252258	c_{52}	−12492018077	c_{74}	39978552
c_9	67919	c_{31}	−2763655418	c_{53}	12718047714	c_{75}	−17446912
c_{10}	−230412	c_{32}	−1105545726	c_{54}	−10549116201	c_{76}	6697504
c_{11}	813530	c_{33}	−312018490	c_{55}	11973045206	c_{77}	−5604576
c_{12}	−825548	c_{34}	7292718779	c_{56}	−9818540634	c_{78}	789472
c_{13}	1919344	c_{35}	−1708673665	c_{57}	7244375881	c_{79}	411456
c_{14}	−7442341	c_{36}	4572159618	c_{58}	−7184734429	c_{80}	94464
c_{15}	7031365	c_{37}	−13772905345	c_{59}	4878378714	c_{81}	318080
c_{16}	−10393589	c_{38}	6070585196	c_{60}	−3132222947	c_{82}	−215808
c_{17}	45861770	c_{39}	−10032469559	c_{61}	2732183900	c_{83}	33024
c_{18}	−42807461	c_{40}	18850529502	c_{62}	−1315418977	c_{84}	−33792
c_{19}	37232139	c_{41}	−8870949775	c_{63}	614276807	c_{85}	13312
c_{20}	−196275440	c_{42}	12219107416	c_{64}	−451575073		
c_{21}	183711765	c_{43}	−17347390633	c_{65}	−106350642		
c_{22}	−77997837	c_{44}	6329537512	c_{66}	200998902		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 90$, and it is the recurrence OEIS A316286 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 5$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{85} - c_1x^{85-1} - \dots - c_{85}$. Its constant term is $-c_{85} = -13312$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 85 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 85, so $\deg q = 85$, and it is monic. It holds from some index t on. From $\max(t, 5)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 85, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 21 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 85, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 13312.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-85 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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